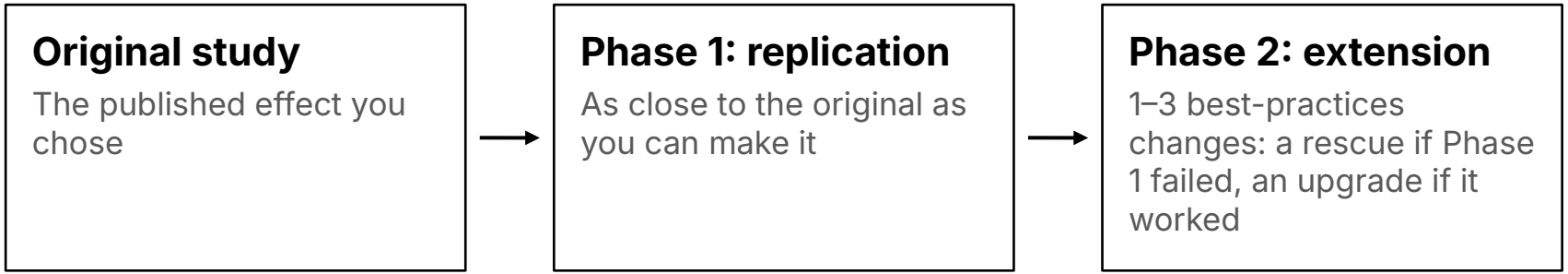
Do best-practices extensions improve replications?

Psych 251 class metascience project, 2026

Proposal and choice points · Monday, November 2

The proposal



The question

Do changes drawn from Experimentology (measurement, design, sample, checks) make a replication more successful, and recover more of the original effect?

Who's in

- Opted-in projects: extra budget, and every opted-in student is an author
- Track 1 only; scientific extensions (Track 2) are excluded
- We expect about 15–20 projects

What we will measure

1 Replication success

Two TAs independently rate each phase against the original, from 0 to 1. A score of .75 or more counts as a success.

2 Effect size ratio

Your effect divided by the original effect, both as standardized mean differences, for Phase 1 and Phase 2. We compute it by re-running your analysis on the data in your repository.

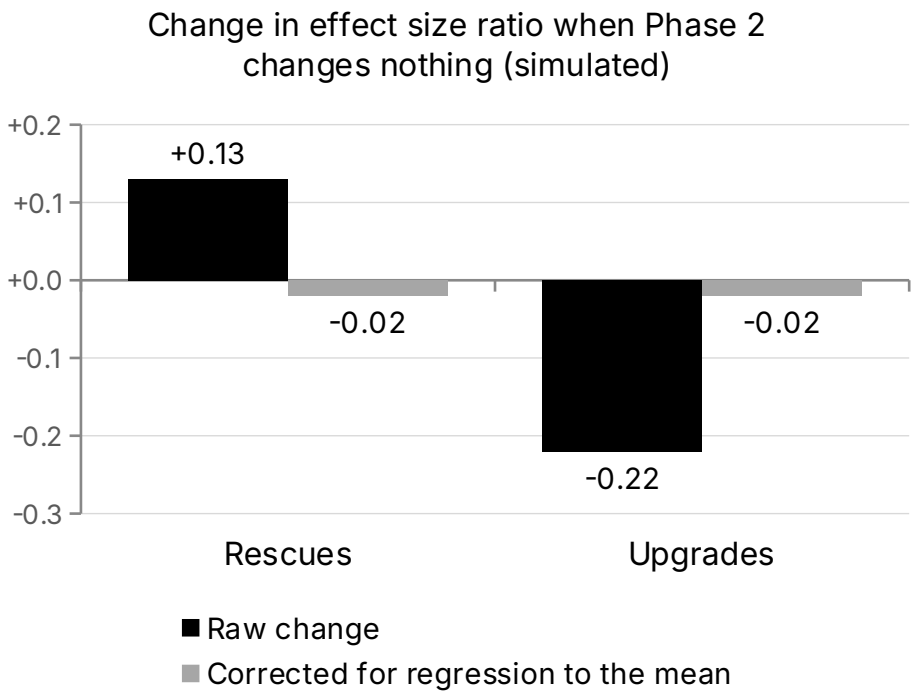
Also: precision (how much the standard error shrank, beyond what sample size alone explains).

5 of 17

rescues in Boyce et al. (2024) mostly or fully replicated (29%)

Different students, timeline, budget and tools, so our comparison with 2024 is descriptive only.

Two things to know about this design



Regression to the mean

Rescues are chosen for low Phase 1 results, so they tend to rise anyway, and upgrades tend to fall. The main test pools all projects, where this cancels; within type we correct for it.

Modest power

With about 20 projects, a 30% larger effect in Phase 2 (say, 65% of the original instead of 50%) is detected only 39% of the time. We report an estimate with an interval, not just yes or no.

Choice points

When do TAs rate Phase 1?

Proposed: at the midterms this week, before any Phase 2 data exist.

Rescue or upgrade: who decides?

Proposed: the TA rating of Phase 1 (.75 or more = upgrade); your declared track is recorded too.

What if Phase 2 changes the design?

Between → within changes the metric. Proposed: recompute on the Phase 1 scale and flag it.

How ambiguous was the original?

Proposed: count differences and unstated choices in your report, plus an agent count a person checks.

Agents checking the original papers

Proposed: an agent flags statistical inconsistencies; a TA verifies each flag after Phase 2 data are in.

Studying us, not the experiments

Attitudes, agent logs, quizzes: need a new IRB protocol and committee sign-off. Opt-in, never graded.

Plan registered on OSF by Nov 13 · Phase 2 data collected the week of Nov 16